BFS Traversal & Distances

Graphs, Breadth-First Search, Shortest Paths · Mentor-Led Lab (Student Edition)

Developed by TalenciaGlobal

Difficulty ★★ Intermediate

Problem

Run a breadth-first search from a source vertex. Report the BFS visiting order and the shortest hop distance from the source to every vertex (-1 if unreachable).

Implement BFS with a queue from scratch; in an unweighted graph it yields shortest paths.

What to Compute

- BFS Order: the order vertices are first visited from the source.
- Distances: the shortest hop distance to each vertex 0..N-1 (the source is 0, unreachable is -1).

Input / Output Format

Input: Line 1: N M. Next M lines: u v. Then a line with the source S.

Output: Two lines: BFS Order and Distances.

Examples

Example 1

Input: 6 5 0 1 0 2 1 3 2 3 3 4 0

Output:

```
BFS Order: 0 1 2 3 4
Distances: 0 1 1 2 3 -1
```

Explanation: From 0, vertices 1 and 2 are one hop away, 3 is two hops, 4 is three; vertex 5 has no edges, so its distance is -1.

Example 2

Input: 4 3 0 1 1 2 2 3 0

Output:

```
BFS Order: 0 1 2 3
Distances: 0 1 2 3
```

Explanation: A path graph: each vertex is one hop farther than the previous.

Constraints

- BFS explores neighbours in sorted (ascending) order.
- Distances are in edges (hops); the source is 0.
- Unreachable vertices have distance -1.

Sample Run

Sample Input

```
6 5
0 1
0 2
1 3
2 3
3 4
0
```

Sample Output

```
BFS Order: 0 1 2 3 4
Distances: 0 1 1 2 3 -1
```